

Keys in DBMS

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What is a Key?

- A key is an attribute or set of attributes used to uniquely identify a record (row) in a table.
- Why keys are important:
 - ✓ Ensure uniqueness
 - ✓ Maintain data integrity
 - ✓ Establish relationships between tables

Types of Keys

- **Super Key**
- **Candidate Key**
- **Primary Key**
- **Alternate Key**
- **Foreign Key**

Super Key

- It is a combination of all possible attributes that can uniquely identify the tuple in a given relation
- It is a superset of candidate key
- A table can have many super keys
- It may contain attributes that are redundant
- It may include extra attributes that aren't important for uniqueness but still uniquely identify the row.

Example: Super Key

Some Super Keys

- {Roll_No, Aadhar_No}
- {Aadhar_No, Email_ID}
- {Roll_No, Name}
- {Roll_No, Branch}
- {Roll_No, Name, Branch}

Roll_No	Name	Aadhar_No	Email_ID	Branch
2K21CO01	Mohan	88213211	m@gmail.com	CSE
2K21EC02	Rohan	28341212	r@gmail.com	EC
2K21IT03	Abraham	65762231	a@gmail.com	IT
2K21SW04	Vernika	98322212	v@gmail.com	SWE

Relation: Student

Candidate Key

- **A minimal super key with no extra attributes.**
- It is a minimal super key with no redundant attributes
- It is a distinct set of attributes from which primary key can be selected
- All candidate keys are super keys

Example 1: Candidate Key

Candidate keys possible

- {Roll_No}
- {Aadhar_No}
- {Email_ID}

Roll_No	Name	Aadhar_No	Email_ID	Branch
2K21CO01	Mohan	88213211	m@gmail.com	CSE
2K21EC02	Rohan	28341212	r@gmail.com	EC
2K21IT03	Abraham	65762231	a@gmail.com	IT
2K21SW04	Vernika	98322212	v@gmail.com	SWE

Relation: Student

EXAMPLE 2: Candidate Key

Table: STUDENT_COURSE

STUD_NO	TEACHER_NO	COURSE_NO
1	001	C001
2	056	C005

A composite candidate key example: {STUD_NO, COURSE_NO} can be a candidate key for a STUDENT_COURSE table.

Primary Key

- A candidate key selected to uniquely identify a record/tuple.
- Properties:
 - Unique
 - NOT NULL
 - Only one per table

Example: Primary Key

Primary key
{Roll_No}

Roll_No	Name	Aadhar_No	Email_ID	Branch
2K21CO01	Mohan	88213211	m@gmail.com	CSE
2K21EC02	Rohan	28341212	r@gmail.com	EC
2K21IT03	Abraham	65762231	a@gmail.com	IT
2K21SW04	Vernika	98322212	v@gmail.com	SWE

Relation: Student

Alternate Key

- **Candidate keys not chosen as primary key.**
- Act as backup identifiers

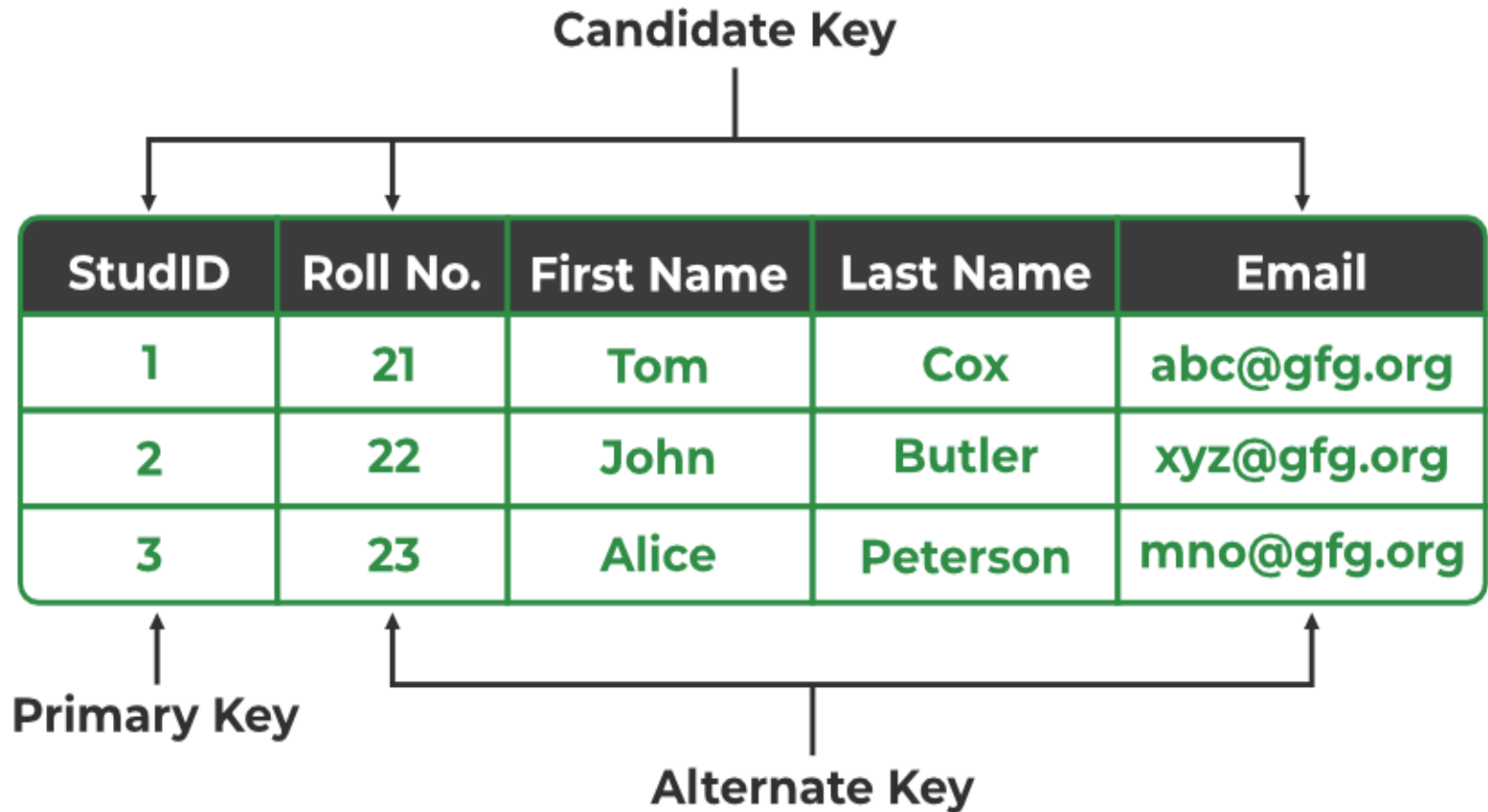
Example: Alternate Key

Alternate keys

- {Aadhar_No}
- {Email_ID}

Roll_No	Name	Aadhar_No	Email_ID	Branch
2K21CO01	Mohan	88213211	m@gmail.com	CSE
2K21EC02	Rohan	28341212	r@gmail.com	EC
2K21IT03	Abraham	65762231	a@gmail.com	IT
2K21SW04	Vernika	98322212	v@gmail.com	SWE

Relation: Student



Foreign Key

- A key used to link two tables together
- When Table B refers Primary Key of Table A, then the Primary Key attribute Table A will be added into Table B as Foreign Key.
- Foreign key can be NULL.
- Foreign Key ensures referential integrity.

Example 1: Foreign Key

Roll_No	Name	Aadhar_No	Email_ID	Branch
2K21CO01	Mohan	88213211	m@gmail.com	CSE
2K21EC02	Rohan	28341212	r@gmail.com	EC
2K21IT03	Abraham	65762231	a@gmail.com	IT
2K21SW04	Vernika	98322212	v@gmail.com	SWE

Relation: Student

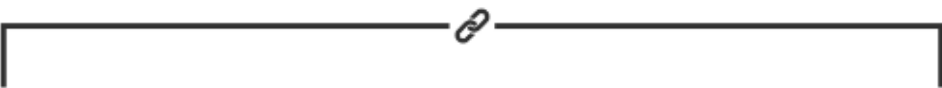
Course_ID	Course_Name	Roll_No
CO201	COA	2K21CO01
CO202	DBMS	2K21EC02
CO203	OS	2K21IT03

Relation: Course_Enrolled

Primary Key of *relation Student* will act as **Foreign Key** in *relation Course_Enrolled*

Primary Key

Foreign Key



ID	Name	Course
2041	Tom	Java
2204	John	C++
2043	Alice	Python
2032	Bob	Oracle

Student Details

ID	Marks
2041	65
2204	55
2043	73
2032	62

Student Marks

Partial Key (Weak Entity)

- Used to identify weak entities.
- Properties
 - Works with owner entity key
 - Cannot uniquely identify alone

Super Key vs Candidate Key vs Primary Key

Basis	Super Key	Candidate Key	Primary Key
Definition	Any attribute or set of attributes that uniquely identifies a record in a table	A minimal super key that uniquely identifies a record	A candidate key chosen to uniquely identify records
Minimality	Not minimal; may contain extra attributes	Minimal; no attribute can be removed	Minimal
Uniqueness	Yes	Yes	Yes
NULL values	Not allowed	Not allowed	Not allowed
Number allowed in a table	Many	One or more	Only one
Selection	Not selected explicitly	Eligible to be selected as primary key	Selected from candidate keys
Relationship with other keys	Superset of candidate keys	Subset of super keys	Subset of candidate keys
Example	(Roll_No, Name)	Roll_No	Roll_No